



REPUBLIC OF GHANA

**PETROLEUM (EXPLORATION AND
PRODUCTION) (MEASUREMENT)
REGULATIONS, 2016**

(L.I. 2246)

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(MEASUREMENT) REGULATIONS, 2016**

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In exercise of the power conferred on the Minister by section 32 (2) (a) of the Petroleum (Exploration and Production) Act, 1983 (P.N.D.C.L. 84), these Regulations are made this 26th day of July, 2016.

Preliminary Provisions

Application

1. These Regulations apply to the

- (a) planning, design, testing, calibration, operation and maintenance of a metering system, equipment and methods for
 - (i) measuring the quantities of petroleum produced, transported or sold,
 - (ii) determining allocated quantities of petroleum produced, transported or sold, and
 - (iii) measuring of petroleum used for fuel or flare gas in petroleum activities;
- (b) planning, design and operation of an allocation system for the allocation of petroleum between contractors, licensees or the Corporation; and
- (c) documentation, reporting and applications related to the measurement, allocation, metering system, equipment or method of measurement.

Purpose

2. The purpose of these Regulations is to ensure that an accurate measurement and allocation of petroleum forms the basis for the determination of revenue that accrues to the Republic, a contractor, licensee or the Corporation.

Compliance

3. A contractor, licensee or the Corporation shall

- (a) comply with these Regulations, directives and instructions issued by the Commission pursuant to these Regulations;
- (b) ensure that an employee, an agent or a subcontractor of the contractor, licensee or the Corporation complies with these Regulations, directives and instructions issued by the Commission pursuant to these Regulations; and

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- (c) ensure that a person acting directly or indirectly for the contractor, licensee or the Corporation complies with these Regulations, directives and instructions issued by the Commission pursuant to these Regulations.

Supervision and inspection

- 4. (1) The Commission may
 - (a) conduct the supervision and inspection of a metering system and an allocation system from the design stage to the operation stage; and
 - (b) in consultation with the Standards Authority, appoint an independent person to conduct supervision and inspection on a metering system or an allocation system from the design stage to the operation stage.
- (2) A contractor, licensee or the Corporation whose metering system and allocation system are the subject of supervision or inspection, shall assist the Commission or the person appointed by the Commission to carry out the supervision or inspection.
- (3) The Commission may require a contractor, licensee or the Corporation to bear the expenses of the supervision or inspection.

Placement of seals on critical valves

- 5. An authorised agency may place a seal on the export valve downstream of a metering station to prevent offloading of petroleum without authorisation.

Use of standards

- 6. (1) The Commission may approve of the use of a technology or method other than that specified in these Regulations and in standards recommended by the Commission where
 - (a) the contractor, licensee or the Corporation demonstrates that the standard, technology or method employed achieves a similar or better level of fidelity, uncertainty and reliability than that specified in these Regulations; or
 - (b) the criterion for the development, testing and operation of the technology or method is established.

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(2) The Commission may approve simplifications in metering if supported by a risk-cost-benefit analysis as set out in the Third Schedule.

Requirements Relating to Management System

Quality management system

7. (1) A contractor, licensee or the Corporation shall develop, implement and maintain a quality management system for petroleum measurement and allocation in accordance with internationally recognised standards.

(2) The quality management system shall form part of the overall management system of a contractor, licensee or the Corporation.

(3) The contractor, licensee or the Corporation shall inform the Commission of a revision of the quality management system.

Responsibility and qualifications

8. A contractor, licensee or the Corporation shall

- (a) nominate a person with the requisite technical competence to be responsible for the metering systems and the allocation systems;
- (b) inform the Commission of the person nominated under paragraph (a);
- (c) specify the responsibility for the on-site day-to-day operation of the metering system; and
- (d) ensure that a person engaged in fiscal metering activities possesses the requisite qualifications and competence to perform the activities.

General Requirements Relating to Allocation, Measuring and Metering System

Uncertainty requirements

9. (1) The maximum allowable measurement uncertainty limits at ninety-five percent confidence level are as set out in the First Schedule.

(2) A contractor, licensee or the Corporation shall carry out an uncertainty analysis for a metering system in accordance with internationally recognised standards with the resulting uncertainty presented at ninety-five percent confidence level.

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(3) The contractor, licensee or the Corporation shall ensure that the uncertainty analysis includes sensitivities of all significant uncertainty contributors to the overall uncertainty and covariances between the uncertainty contributors where applicable.

(4) The maximum uncertainty limits at ninety-five percent confidence level related to a metering system are as set out in the Second Schedule.

(5) The contractor, licensee or the Corporation shall take reasonable steps to avoid systematic measurement errors.

Calibration and traceability

10. (1) A contractor, licensee or the Corporation shall ensure that measurements and calibrations made under these Regulations are traceable to national and international measurement standards.

(2) The contractor, licensee or the Corporation shall ensure that an instrument used for calibration or tracing is adjusted where a calibration result has significant influence on a fiscal figure.

(3) A third party laboratory used to calibrate equipment covered by these Regulations shall

(a) be accredited in accordance with internationally recognised standards; and

(b) have a documented uncertainty equal or better than those set out in the Second Schedule.

(4) The contractor, licensee or the Corporation shall ensure that a laboratory analysis of the composition or quality of petroleum for custody transfer and allocation purposes is carried out by a competent or an accredited laboratory.

Units of measurement

11. (1) The readings of

(a) a metering system shall be in SI units;

(b) pressure may be in bar; and

(c) temperature may be in degree Celsius.

(2) The critical parameters of a metering system shall be measured in SI units.

(3) A contractor, licensee or the Corporation shall

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- (a) report petroleum measurements to the Commission in SI units; and
- (b) convert and report to the Commission, the net volume of oil in barrels.

Reference conditions

12. (1) The reference conditions for temperature and pressure to determine standard volume is 15 °C and 101.325 kPa.

(2) A reference pressure other than 101.325 kPa may be used for condensate and liquefied petroleum gas.

Bypass of metering system

13. A person shall not bypass a metering system.

Allocation systems

14. (1) A contractor, licensee or the Corporation shall develop an allocation system for the allocation of petroleum quantities commingled from fields consisting of different ownership and royalty interests.

(2) The contractor, licensee or the Corporation shall maintain a field balance or a plant balance of fluids produced, exported, injected, transferred, flared, disposed of or used for fuel, gas lift or other utilities.

(3) In the case of a pipeline system, a contractor, licensee or the Corporation shall maintain a mass balance.

Design of a Metering System

General design considerations

15. (1) A contractor, licensee or the Corporation shall ensure that the design of a metering system, including installation and equipment, are in accordance with these Regulations and internationally recognised standards.

(2) The contractor, licensee or the Corporation shall ensure that

- (a) the metering system is suitable for
 - (i) the relevant type of measuring,
 - (ii) the fluid properties, and
 - (iii) the petroleum quantities to be measured;
- (b) the metering system can measure the full range of planned

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petroleum flow without any component operating outside the specified working range;

- (c) measurement errors that may be caused by similar drifting in duplicated instruments due to process conditions are minimised;
- (d) each part of the metering system is easily accessible for maintenance, inspection and calibration;
- (e) external conditions do not affect the operation and maintenance of the metering system;
- (f) it is possible to safely remove an individual element from a metering system for custody transfer measurements without a shut-down of the production system;
- (g) parameters related to the fiscal calculations are readily accessible in the computer part of the metering system or through a service computer; and
- (h) provision is made for condition monitoring.

Design of mechanical part of metering systems

16. (1) A contractor, licensee or the Corporation shall design the mechanical part of the metering system so that representative measurements are achieved as input signals for a fiscal calculation.

- (2) A single-phase metering system shall be designed
 - (a) to meet the uncertainty requirements set out in the First Schedule; and
 - (b) shall include
 - (i) provisions to maintain all fluids in a single-phase;
 - (ii) sufficient redundancy to measure maximum flow within required measurement uncertainty where a meter or a meter run is out of service;
 - (iii) measures to provide a sufficiently developed flow profile at the position of the measurement;
 - (iv) a temperature measurement point and a pressure measurement point located in a manner that ensures that measurements are representative of conditions at a meter and a proving device;

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- (v) provisions for the verification of temperature measurements;
- (vi) measures to prevent ambient temperature and other external factors affecting a measurement; and
- (vii) adequate valve arrangements for operations and maintenance.

(3) A metering system for sales and custody transfer of petroleum shall be in accordance with nationally and internationally recognised standards for custody transfer measurements.

(4) A metering system for sales and custody transfer of liquid petroleum

(a) shall be designed to meet the requirements set out in subregulation (2); and

(b) shall include

- (i) a displacement prover for the calibration of duty meters; or
- (ii) a master meter in a by-pass loop with provisions for flushing;
- (iii) a provision for the calibration of the proving device at the place of operation; and
- (iv) a provision for the injection of inhibitors where the metering system is exposed to deposits.

(5) A metering system for sales and custody transfer of gas shall be designed to meet the requirements in subregulation (2).

(6) A fuel gas metering system shall be designed to meet the requirements in subregulation (2), where the requirement in subparagraph (ii) of paragraph (b) of subregulation (2) may be fulfilled by

- (a) a single meter run and a by-pass run with sufficient measures to prevent accidental flow through the by-pass run; or
- (b) a single meter run with two meters in series.

(7) A flare gas metering system shall be designed

- (a) in accordance with paragraph (a) of subregulation (2) and subparagraphs (iii), (iv), (v) and (vi) of paragraph (b) of subregulation (2); and

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- (b) to ensure that the sensors of the flare gas metering system are accessible for maintenance during operation.
- (8) A single-phase metering system and a multi-phase metering system forming a part of an allocation system shall be designed to provide a sound basis as input to the allocation system.
- (9) A valve that has an impact on a measurement shall
 - (a) have double barriers;
 - (b) have a monitoring function;
 - (c) be accessible for inspection; and
 - (d) not generate noise that interferes with the signal of an instrument.

Design of instrument parts

17. (1) A contractor, licensee or the Corporation shall ensure that the instrument part of a metering system

- (a) has specifications that are consistent with the requirements as set out in the Second Schedule;
 - (b) consists of a material compatible with the fluid that it is exposed to;
 - (c) is located in a manner that achieves representative measurements as an input signal for fiscal calculation;
 - (d) is mounted according to the instructions of the manufacturer and recommendations made in national or internationally recognised standards recommended by the Commission;
 - (e) includes a signal transmission system which provides fidelity and security of an instrument signal; and
 - (f) facilitates the active interpretation of diagnostic data, where applicable.
- (2) A flow meter shall
- (a) be suitable for the intended application with regard to process parameters including
 - (i) maximum and minimum flow rates,
 - (ii) Reynolds number,
 - (iii) temperature,

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- (iv) pressure,
 - (v) density, and
 - (vi) viscosity;
- (b) have a dimension that ensures that the flow meter is calibrated over the operating range; and
- (c) where applicable,
 - (i) have documented performance with respect to fractions of other fluid phases, that is water, gas or liquids, than the one subject to the measurement;
 - (ii) incorporate temperature and pressure compensation routines;
 - (iii) have provisions for software identification; and
 - (iv) have provisions for protection of software and data.
- (3) A displacement prover shall
 - (a) be sized for proving of a flow meter over its operating range;
 - (b) be equipped with piping for calibration, ventilation, drainage, and where necessary, pressure relief; and
 - (c) in the case of a sphere displacer be
 - (i) bidirectional; and
 - (ii) equipped with switches for at least four distinct volumes; and
 - (d) in the case of a piston displacer be mounted vertically, where practicable.
- (4) A master meter shall
 - (a) meet the requirements as stipulated for a liquid flow meter in the Second Schedule;
 - (b) be sized to prove a line meter such that the operating range of the line meter falls within the proven operating range of the master meter; and
 - (c) be selected to minimize the effect of variances in flow rate and viscosity.
- (5) A temperature measuring instrument, a pressure measuring instrument, a differential pressure measuring instrument and a density measuring instrument shall each

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- (a) transmit a signal to a computer part digitally;
 - (b) have an integrated local indicator displaying the measurement result; and
 - (c) provide for sensor linearization, where applicable.
- (6) A density measuring system of by-pass type shall
- (a) facilitate corrections to the conditions at the point of flow measurement;
 - (b) be equipped with two density measuring instruments; and
 - (c) be configured in a manner that allows the density measuring instruments run in parallel and in series.
- (7) A gas chromatograph shall be capable of analysing all components within the uncertainty limits as set out in the Second Schedule.
- (8) A single-phase gas metering system for custody transfer shall be equipped with two independent gas analysis systems.
- (9) A gas chromatograph system shall be designed to determine a flow-weighted calorific value.

Design of sampling system

18. (1) A single-phase metering system shall include a sampling system for determining the representative mean composition of the bulk quantity of petroleum and quality of the bulk quantity of petroleum.

- (2) A sampling system for a single-phase metering system shall
- (a) be capable of providing a sample that, at the location and time of samplings has the same composition as the average composition of the petroleum over the whole cross-section of the pipeline;
 - (b) be capable of transporting a sample to a storage unit or an analytic unit without
 - (i) loss of liquid, solid or gas; and
 - (ii) contamination of the sample;
 - (c) be configured in a manner that sampling is automatic;
 - (d) be configured in a manner that sampling is flow proportional;

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- (e) be capable of collecting representative sample volumes;
 - (f) be capable of dividing a sample into a number of sub-samples in a manner that ensures that each sub-sample has the same composition as the sample; and
 - (g) include a facility for manual sampling.
- (3) A gas sample may be transferred directly to an analytical unit.
- (4) Where a gas sample is transferred directly to an analytical unit, paragraphs (a), (b) and (c) of subregulation (2) shall apply.

Design of computer part

19. (1) A metering system shall include a dedicated computer part for purposes of

- (a) calculating flow of petroleum;
 - (b) calculating the quantities of petroleum accumulated;
 - (c) reporting; and
 - (d) system control.
- (2) A contractor, licensee or the Corporation shall
- (a) dedicate a flow computer to each meter run; or
 - (b) where a multiple meter run is computed by one machine, provide a hot operating independent standby to allow maintenance or replacement of parts of a computer system to be carried out without interruption to the measurement of petroleum.
- (3) The contractor, licensee or the Corporation shall ensure that the computer part under subregulation (1)
- (a) is capable of performing calculations within the uncertainties as set out in the Second Schedule;
 - (b) is capable of performing a full calculation cycle within one second;
 - (c) is equipped with security functions to ensure that measurement figures cannot be changed as a result of incidents of a technical nature or as a result of a manual fault;
 - (d) is capable of documenting the various fiscal parameters and the fiscal quantities calculated;

